CogNarr Story Graph

A Formal Semantics for Narrative as a Belief Model

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Active Inference Institute

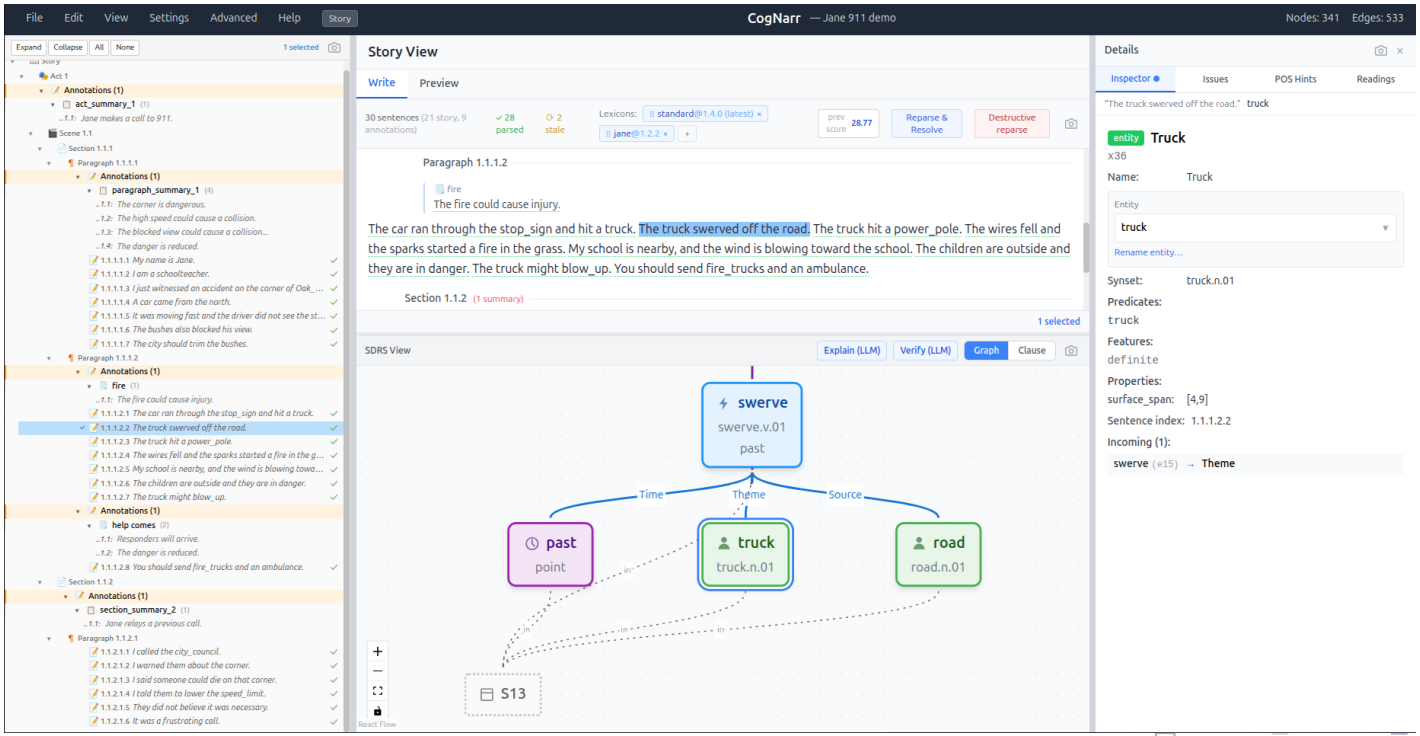
Applied Active Inference Symposium, November 2026
with panelists Glen EP Ropella and Michael Lennon
cognarr.org

Today's Talk

- World premier of the CogNarr app. Years in the making, now close to open-source release on GitHub. Paper coming.
- Today's plan: several framing slides, a short live demo of the application, a panel discussion, then more of the application with questions and answers.
- An open invitation. We are seeking collaborators, use cases, projects, and ideas. Visit cognarr.org, and reach us from there.

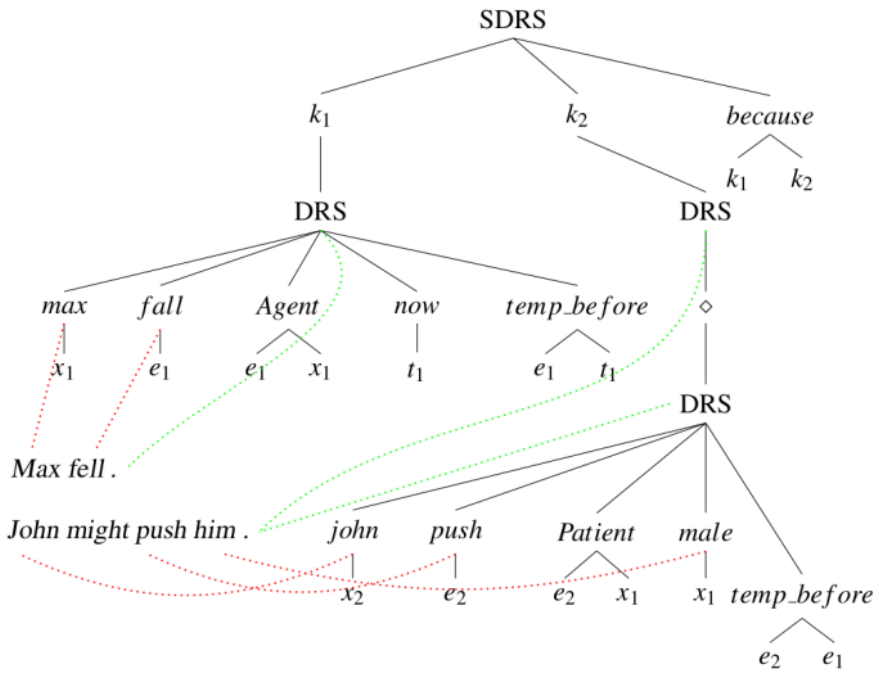
What Is CogNarr?

- CogNarr (Cognitive Narrative) is the implementation of the story graph meaning representation (Boik, 2024a, 2024b).
- It reads a narrative and builds an explicit, interactive, formal semantic graph a person can inspect and correct.
- It's the core service of the wider CogNarr Ecosystem, an Active Inference Institute project for facilitating group cognition at scale.



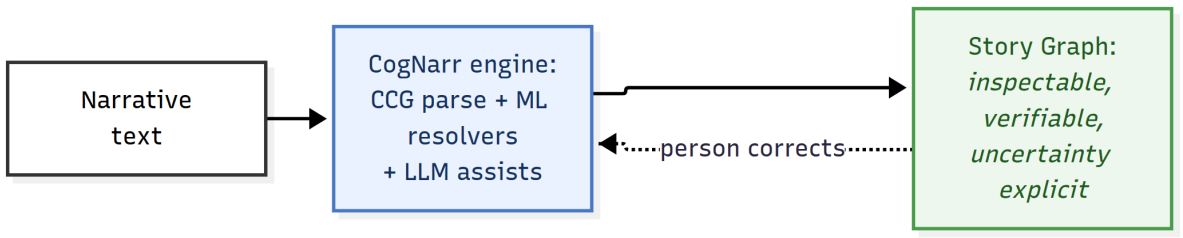
What Is a Meaning Representation?

- A meaning representation makes the meaning of a sentence explicit, allowing both people and computers to read and understand it. Example: a Discourse Representation Tree Structure for "Max fell. John might have pushed him."
- A CogNarr story graph has much more, including a narrative timeline, argumentation structure, explicit uncertainty, and co-resolved senses, identity, and coreference across the whole queryable graph.



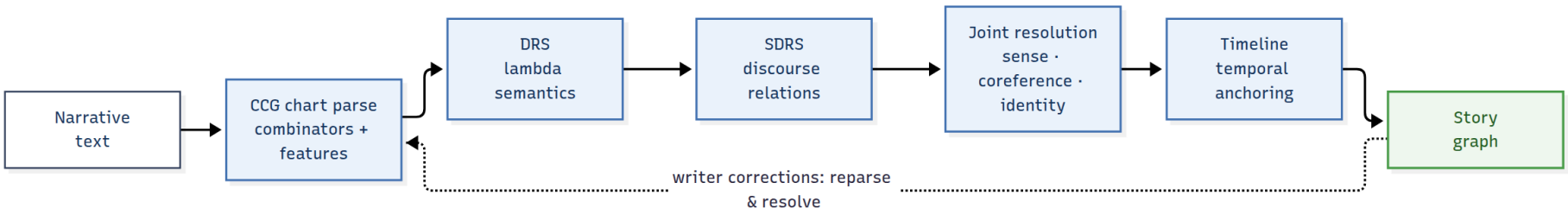
Why Not Just an LLM?

- A bare LLM returns text. You can't directly inspect what it understands about a narrative, how it reasons and predicts, or how it weighs uncertainty.
- CogNarr combines formal semantics, machine learning, and LLM assistance to build a verifiable, symbolic, interactive, semantic graph. The LLM is an assistant, not the whole system.
- Contrastive logical analysis could be performed over large sets of story graphs, more narrative than would fit in a single LLM prompt.



Under the Hood

- Implemented in Julia. Combinatory Categorial Grammar (CCG) semantic parser, interactive story-graph editor.
- The CCG chart parser uses standard combinators over categories with feature unification. That is, it's a principled, rule-based parser that builds meaning compositionally from the words.
- Resources: Open English WordNet (117,953 nouns and 21,479 adjectives to synsets, with hypernym chains) and VerbNet (4,580 verbs for senses and thematic roles).

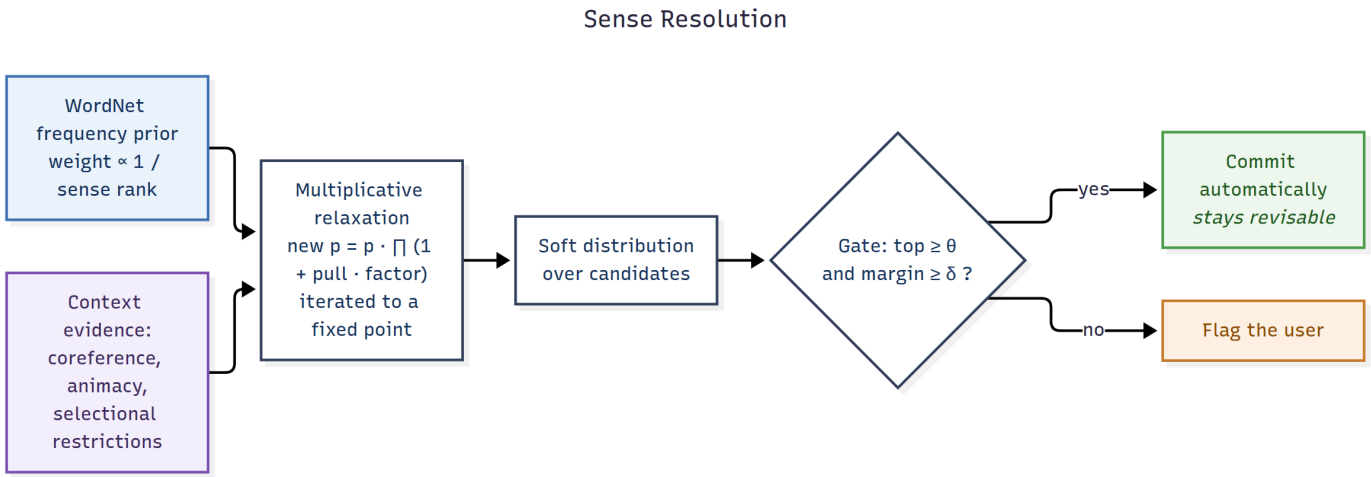


Live Demo

A short tour of CogNarr

How the Soft Score Is Computed

- For each uncertain decision, sense, coreference, and identity, CogNarr keeps a soft distribution over candidates.
- A sense resolution example is shown below.
- A WordNet frequency prior is reweighted by context evidence, and a score-and-margin gate then commits automatically or flags the user.
- A similar process resolves identity and coreference. Soft scores are used elsewhere too, such as in identifying relations-graph motifs.
- Active inference could replace this relaxation, minimizing variational free energy as resolution runs.

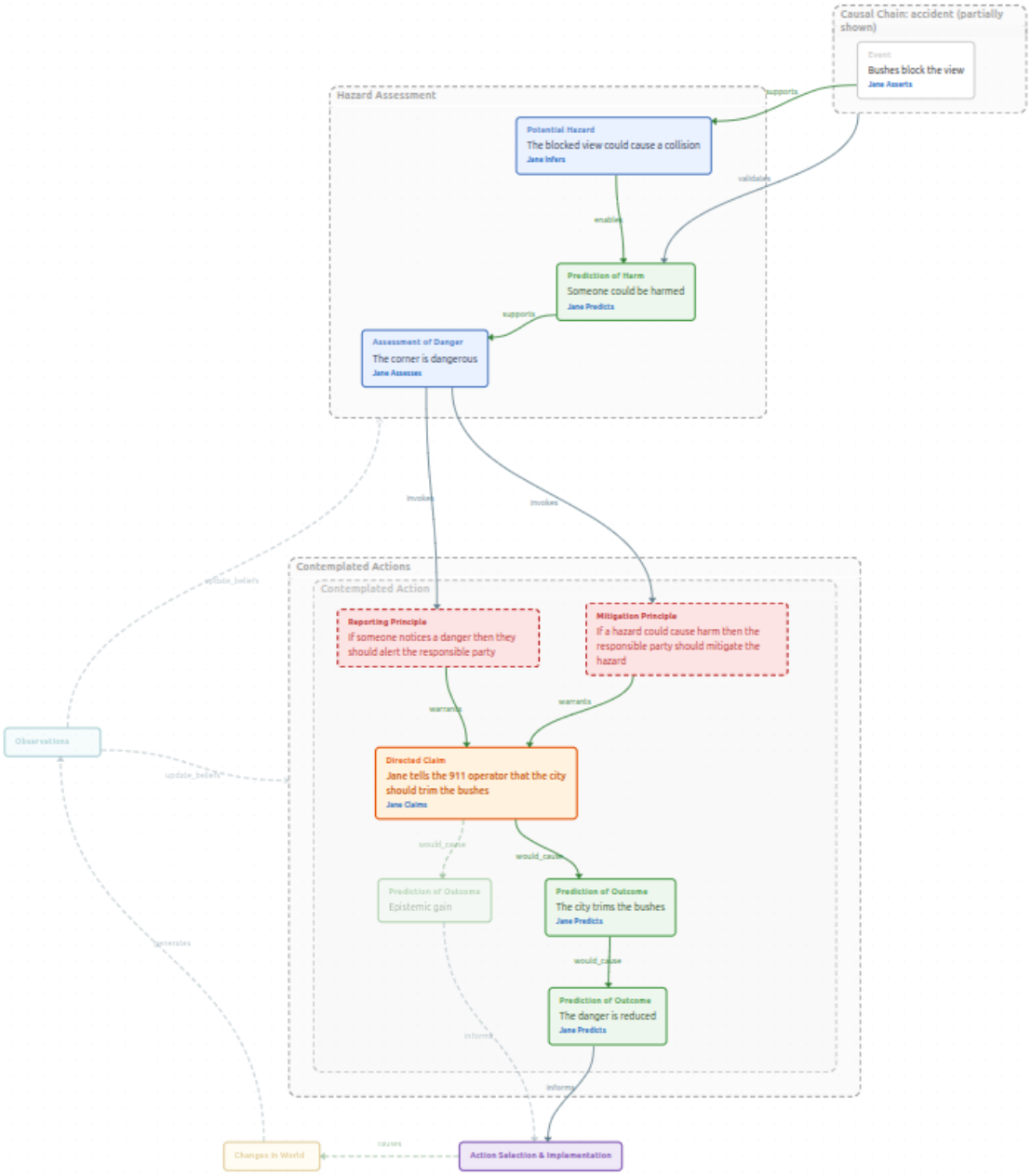


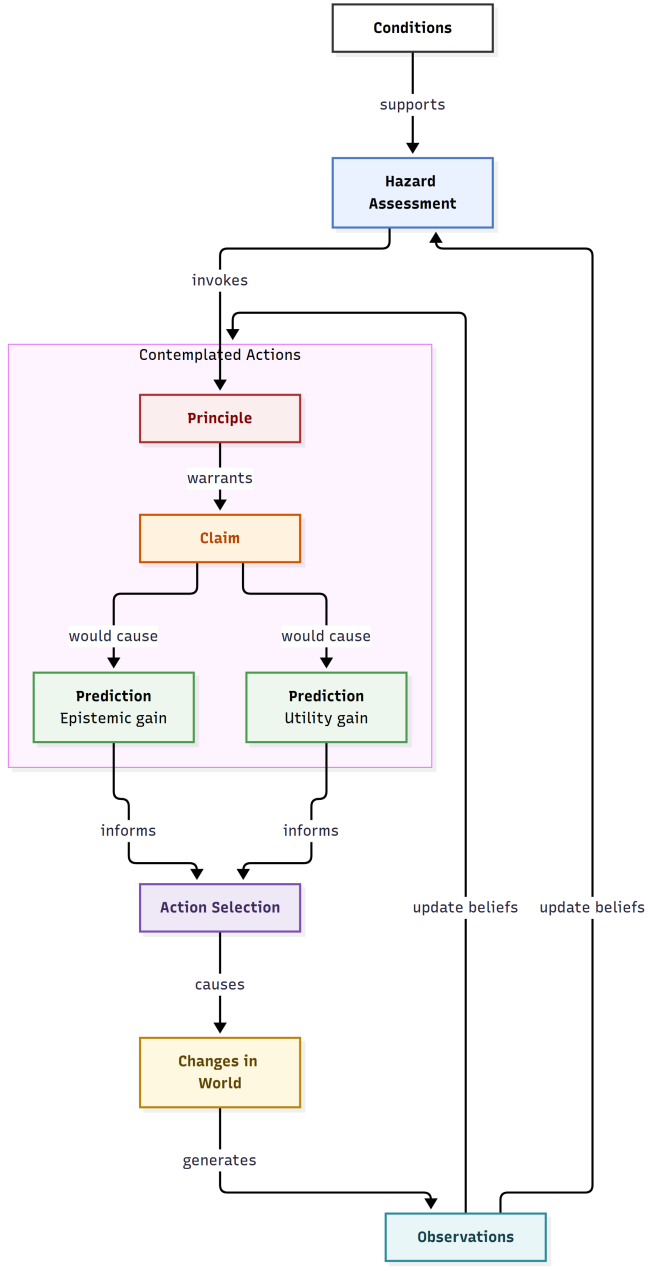
Evaluation

- Parallel Meaning Bank (PMB) parsing (held-out test):
  - v3 $\approx$ 0.71 F1 (clause-level precision and recall)
  - v5 $\approx$ 0.60 SMATCH (whole-graph overlap with gold)
- Ships with a corpus of 40–50 short, fully-resolved stories:
  - A seed for new, narrative-level testing and evaluation, beyond the sentence-level benchmarks above.

CogNarr and Active Inference

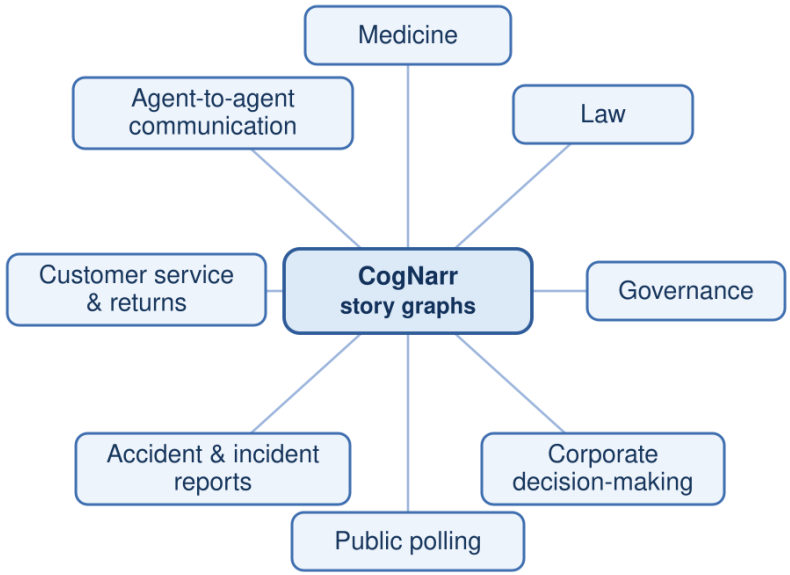
- A story graph represents an instantiated belief model, with uncertainty made explicit.
- A narrative can express the observe $\rightarrow$ learn $\rightarrow$ predict $\rightarrow$ act process, as a human or agent makes decisions using formal active inference, or active inference in the wild.
- Active inference could be used in the internal soft-resolution processes (e.g., sense selection).


Causal Chain: accident (partially shown)
Event
Bushes block the view
Jane Asserts
supports
Hazard Assessment
Potential Hazard
The blocked view could cause a collision
Jane Infers
validates
enables
Prediction of Harm
Someone could be harmed
Jane Predicts
supports
Assessment of Danger
The corner is dangerous
Jane Assesses
invokes
invokes
update_beliefs
Contemplated Actions
Contemplated Action
Reporting Principle
If someone notices a danger then they should alert the responsible party
Mitigation Principle
If a hazard could cause harm then the responsible party should mitigate the hazard
warrants
warrants
Observations
update_beliefs
Directed Claim
Jane tells the 911 operator that the city should trim the bushes
Jane Claims
would_cause
would_cause
Prediction of Outcome
Epistemic gain
Prediction of Outcome
The city trims the bushes
Jane Predicts
generates
would_cause
Prediction of Outcome
The danger is reduced
Jane Predicts
informs
informs
Changes in World
causes
Action Selection & Implementation


Conditions
supports
Hazard Assessment
invokes
Contemplated Actions
Principle
warrants
Claim
would cause
would cause
Prediction
Epistemic gain
Prediction
Utility gain
informs
informs
Action Selection
update beliefs
update beliefs
causes
Changes in World
generates
Observations

Where Can CogNarr Be Used?

- Wherever consequential beliefs are expressed as narrative.
- Agent-to-agent communication over a shared, verifiable structure. Could be a lingua franca for machines.
- The aim behind the project: aligning shared belief models in large-group decision-making.



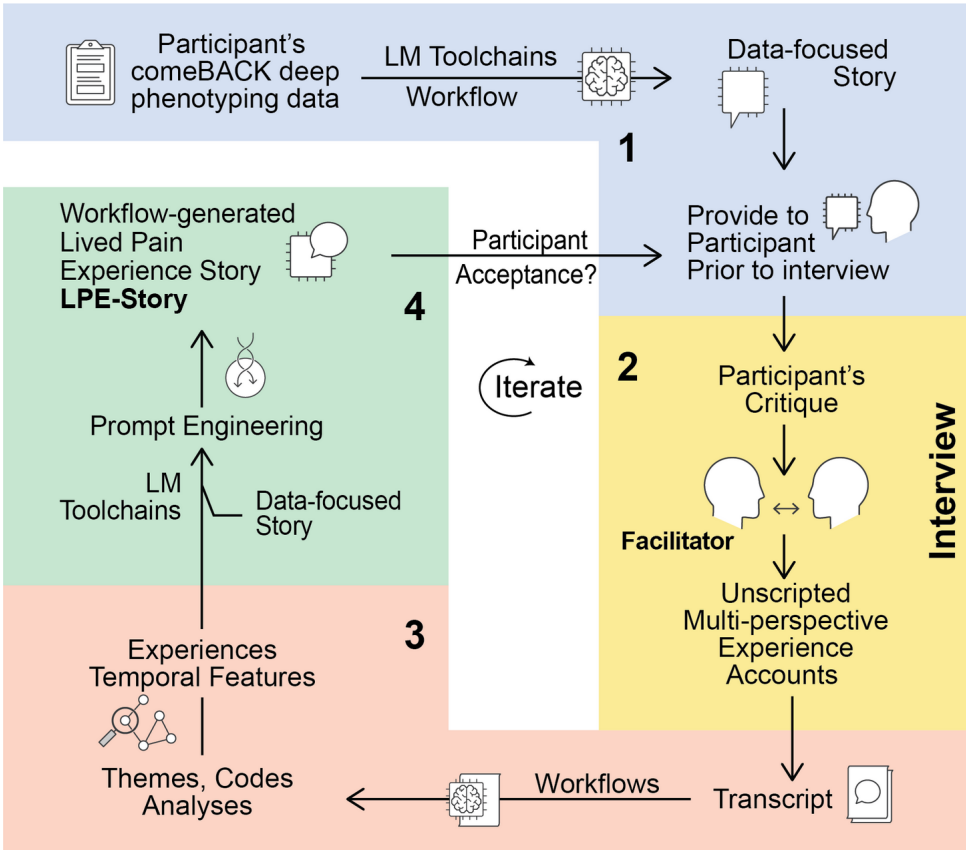
Panel Discussion

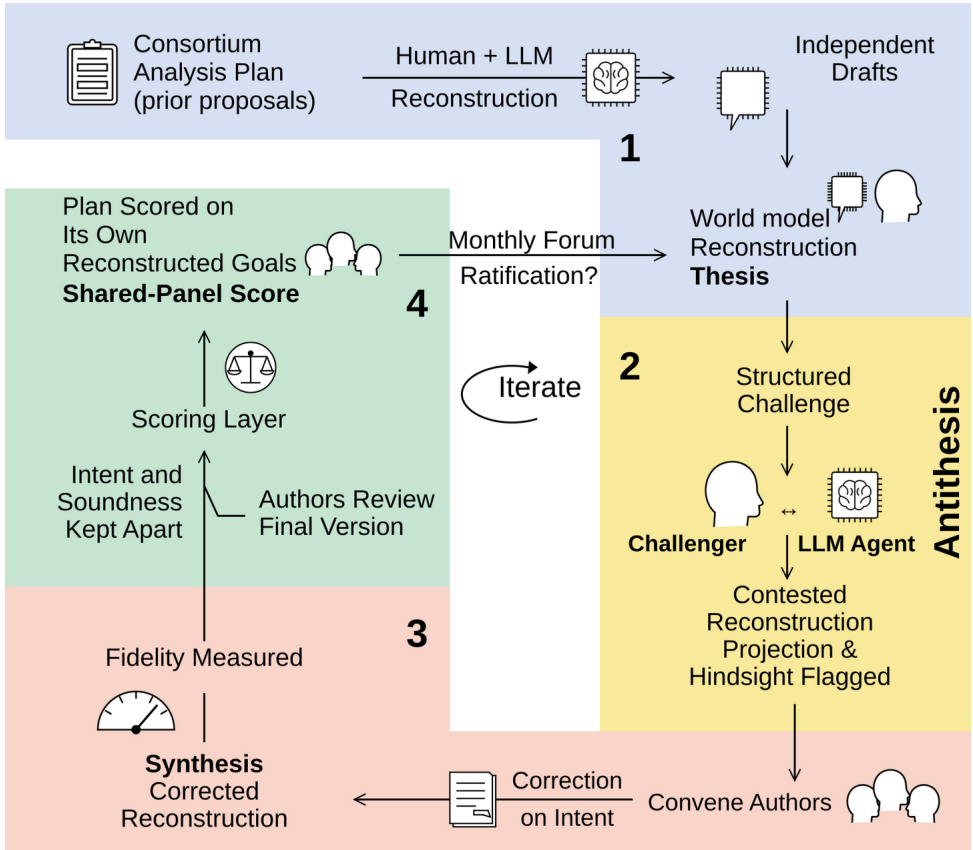
- John Boik — CogNarr and the story graph
- Glen EP Ropella
- Michael Lennon
- Audience — use cases, questions, and ideas

UCSF Planned Uses of CogNarr

Lived Experience (LE) Stories:
Intent is to iterate on a narrative to satisfy both the participant's LE and their medical record.
CogNarr primarily used in Step 4.

Proposals & Analysis Plans:
Intent is to evaluate & iterate on a proposal to satisfy both the proposers' intent and the consortium's objectives.
CogNarr used throughout.





Potential Research Topics

- Alignment of human and/or agent shared beliefs in collective decision making at scale (a sheaf theory question?).
- Improving agent (and robotic) understanding and reasoning, with logical/symbolic validation.
- Improvement of agent-agent communication.
- Assessment of narrative quality (logical consistency, completeness).
- Narrative (and its generation) as a form of evolving, learning belief models.
- A new benchmark: simultaneous entity, coreference, word sense, timing, and argument structure resolution in narrative corpus.
- Extracting causal and argument structure from text.
- Applications in medicine, law, governance, industry, ...

Links to More Information

- CogNarr website: cognarr.org
- Active Inference Institute CogNarr Ecosystem Page
- Boik, JC., 2024a. CogNarr Ecosystem: Facilitating Group Cognition at Scale. ArXiv.
- Boik, JC., 2024b. CogNarr Ecosystem: Preliminary Thoughts on a Story Graph Meaning Representation. OSF.

Live Demo, Continued

A closer look, and your questions

Learn more at cognarr.org